# Amriteshkumar Yadav

**Software Engineer | Fintech & Backend Systems**

Ahmedabad, India | amriteshyadav2004@gmail.com | +91 8320154269 | github.com/Amrit-04 |
linkedin.com/in/amriteshkumar-yadav-214594250

## Education

**SRM Institute of Science and Technology** — Chennai, India
*B.Tech in Computer Science and Engineering (AI/ML Specialization), CGPA: 7.98/10* — *Graduated Jul 2026*

## Technical Skills

**Languages & Web:** Go, Python, C++17, C, SQL, CUDA, Java, TypeScript, Node.js, React.js, Next.js, FastAPI
**Fintech Systems:** FIX 4.2, Kafka, PostgreSQL, Redis (Lua), Event Sourcing, Outbox Pattern, Idempotency
**Quant & ML:** Monte Carlo (VaR/CVaR), HMM, QUBO/QAOA, PyTorch, XGBoost, LightGBM, NumPy
**Testing & Infra:** Rapid (property-based testing), Docker, Kubernetes, AWS, Prometheus, Grafana, Linux

## Experience

**Software Engineering Intern** — Jul 2025 – Jan 2026
*PopFix (now SuckSi) – University-Incubated Nutraceutical Startup* — *Remote*

- Spearheaded **full-stack web development** (**React.js, Next.js**) and cloud setup as the **first technical hire**, configuring production domains, DNS routing, and automated order-processing workflows from scratch.
- Developed and deployed **Python automation** scripts for order tracking, vendor coordination, and logistics workflows, reducing manual processing time by **30%** and scaling operations to **100+ monthly orders**.
- Partnered with founders in go-to-market strategy meetings as the **sole technical voice**, translating business requirements into standardized fulfillment workflows that eliminated manual spreadsheet re-keying.

## Projects

**Tradeflow: Distributed Order Management & Pre-Trade Risk Platform** — GitHub
*Go, PostgreSQL, Redis, Apache Kafka, FIX 4.2 Protocol, Python, Docker, Automated Testing*

- Built an **event-driven order management system (OMS)** in Go (**113 ns/op** state transitions) with PostgreSQL audit logs and a transactional outbox, achieving **14.38 ms median (17.99 ms p99)** end-to-end latency and **zero event loss** across Kafka restarts.
- Engineered an atomic **pre-trade risk engine** in **Redis Lua** using exact integer math to prevent rounding errors, evaluating 7 risk rules (price bands, position caps, kill switch) in **311 ns/op** across 50 concurrent workers.
- Implemented an **idempotent position and P&L tracker** verified across **1,000 randomized property-based tests** (up to 4x duplicate and out-of-order fills), paired with Python scripts for **end-of-day trade reconciliation** and **99% VaR**.

**Moneyball: Quantitative Portfolio Analytics & GPU Risk Simulation Engine** — GitHub
*Python, PyTorch (CUDA), Scikit-learn, XGBoost, LightGBM, Hidden Markov Models (HMM), NumPy*

- Architected a **GPU-accelerated Monte Carlo risk engine** in PyTorch (CUDA) simulating **1,000,000 market scenarios** across 24 assets to compute 99% VaR/CVaR in **29.26 ms**, achieving a **42.9x speedup** over CPU-based NumPy.
- Developed a **market-regime detection pipeline** using a 3-state Hidden Markov Model (HMM) and walk-forward validation, cutting 3-year out-of-sample **max drawdown from 13.58% (1/N baseline) to 8.59% (2.02 Sharpe)**, net of 10 bps turnover costs.
- Designed a **quantum-inspired portfolio optimizer** (14-qubit QAOA/QUBO simulator) in Python, evaluating 16,384 asset combinations to reach **95.49% of the globally optimal risk-adjusted allocation**.

**Real-Time Warehouse Routing & Order-Picking Engine** — GitHub
*C++17, Python, FastAPI, Redis, pybind11, Docker, Locust, Pytest, Prometheus*

- Engineered a **C++17 pathfinding engine** integrated with Python via pybind11, accelerating A* search across a 30,000-cell warehouse grid (**3,294 nodes expanded**) from **15.52 ms to 0.50 ms (31x speedup over Python)**.
- Built a multi-threaded **C++17 multi-stop route optimizer** (Held-Karp dynamic programming and 2-opt search) with cache-friendly memory layouts, cutting 10-stop order-picking latency from **73.57 ms to 1.30 ms (56.5x speedup)**.
- Designed a **FastAPI microservice** and async **Redis caching layer** with O(1) cache invalidation and Prometheus monitoring, reducing **median API latency by 43.8%** at **144 RPS** (**79.7% cache hit rate**; **0% errors**).

## Publications

**IEEE RMKMATE 2026:** *"Workplace Safety Using Computer Vision: A Framework for Personal Protective Equipment Detection and Intrusion Monitoring"* (IEEE Xplore: 11518869 | GitHub) – Co-authored a real-time YOLO and OpenCV pipeline for PPE compliance and restricted-zone intrusion tracking.

## Certifications & Achievements

Quantum Quest Expo 2024 Finalist | Red Hat Certified System Administrator (RHCSA) | Oracle Generative AI Professional (2025) | Oracle OCI Foundations Associate